CVT Guarding

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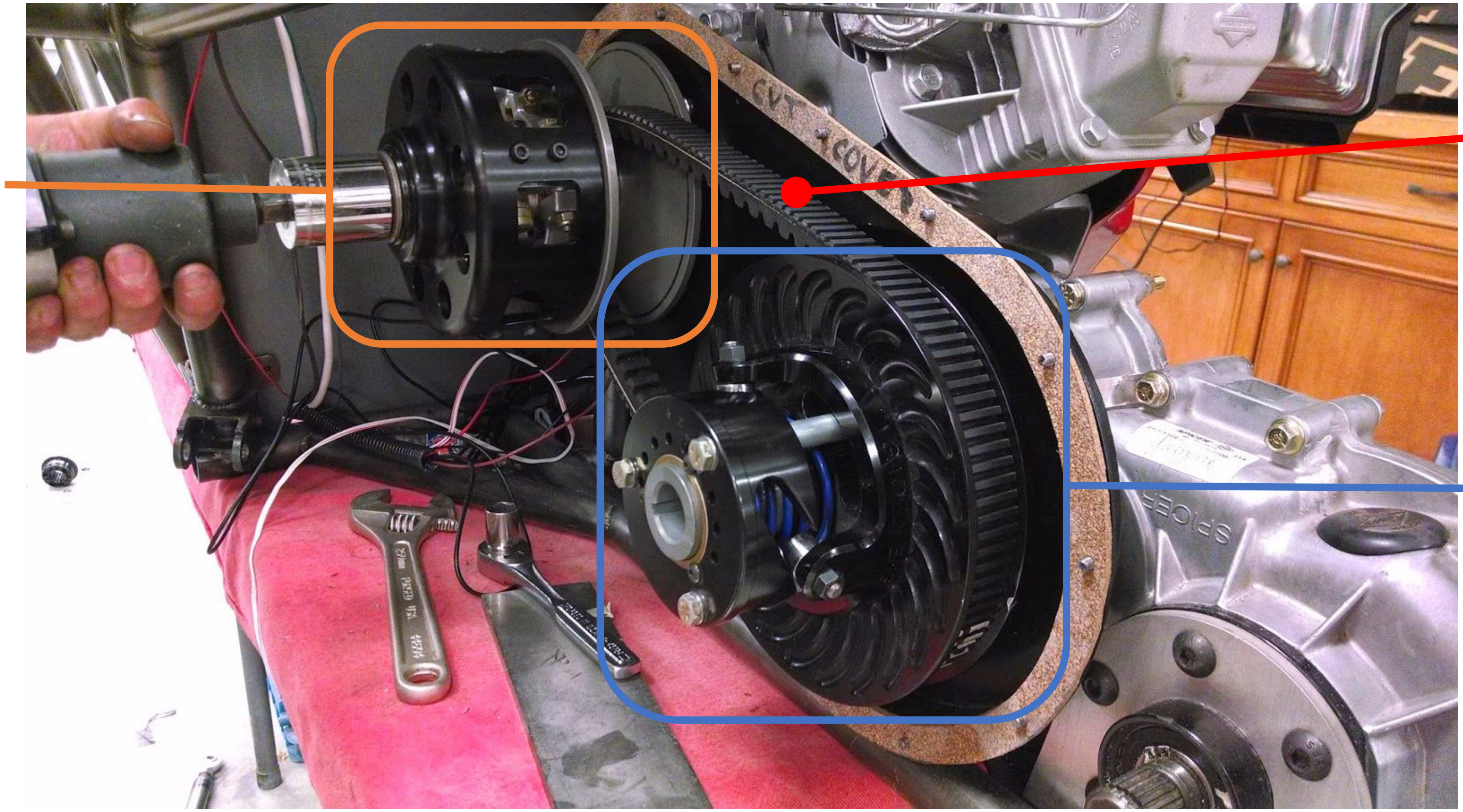
What is the CVT?

Input from engine (diameter increases)



Belt (transmits power between pulleys)

Output to gearbox (diameter decreases)

Credit: UCSB Baja SAE Team

Key Rules & Definitions for CVT Guarding

CVT - Continuously Variable Transmission
HROE - Hazardous Release of Energy - Heavy guarding (RED)
PPAE - Pinch Point and Entanglement - Light Guarding (BLUE)

"HROE guards shall **extend around the entire periphery of the guarded components.**"
HROE guards must be created from "**Steel, at least 1.5 mm (0.06 in.) thick**" or "**Aluminum, at least 3.0 mm (0.12 in.) thick.**"
"The width of the continuous metal band shall be **wider than the entire width of the rotating component by a minimum of 13 mm (0.5 in) on each side.**"
"**A 4 mm (0.25 in) gap between guarding and shaft is permitted.**"
"**Multi-part covers must have at least 0.5 inch (12 mm) of overlap at joints along the periphery.**"
"PPAE guarding for belt, gear, and chain drives shall **cover all directions that HROE guard does not protect.** PPAE guarding **may contain holes or slots for ventilation**. Fastening methods may consist of **threaded fasteners or quick release latches**."
Source: Baja SAE 2025 Rulebook

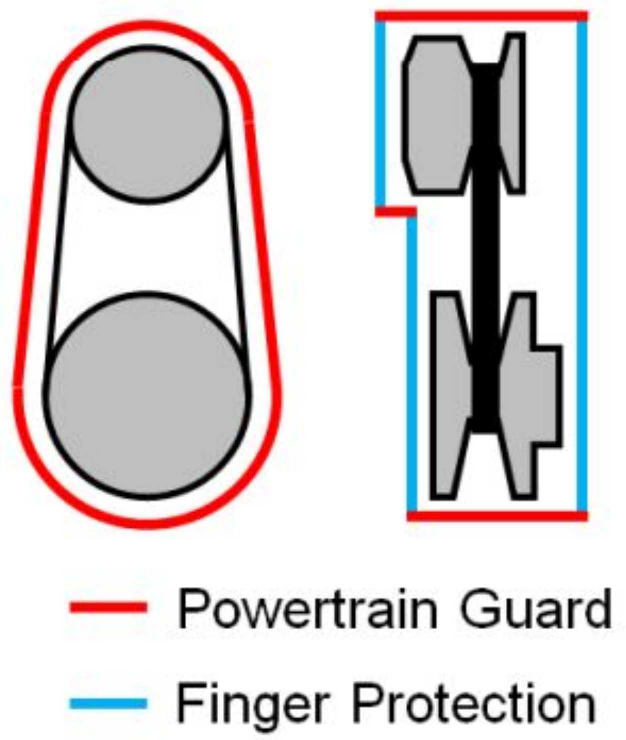
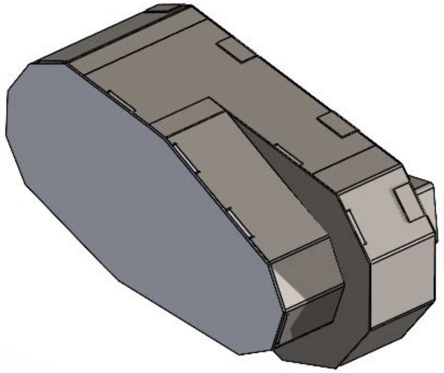


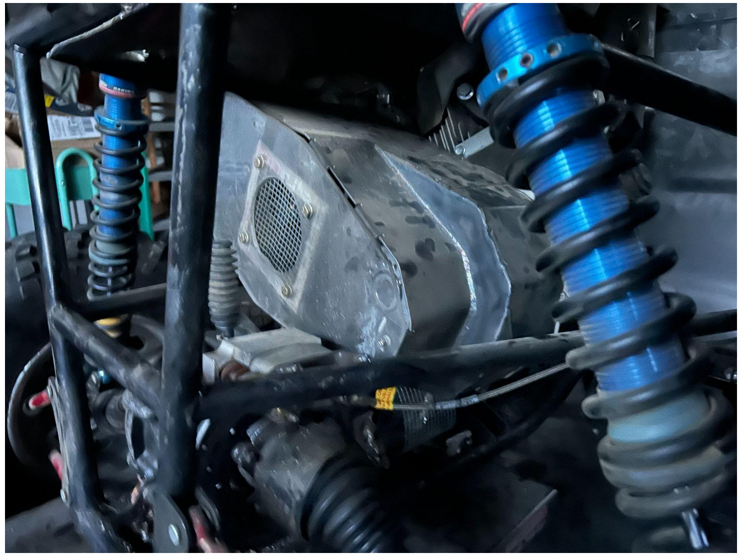
Figure B-54: Powertrain Guard Extent example on a CVT

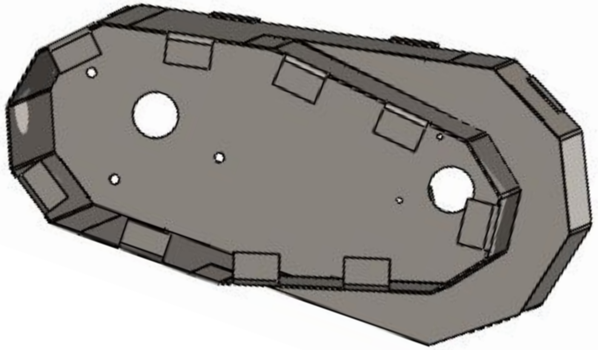
Overview of Previous Design

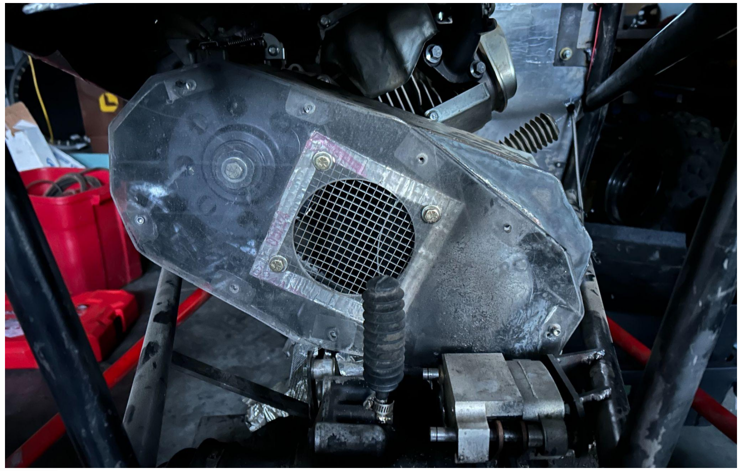
Cons:

- Too hard to take out
- Chunky
- No mounting on the lower end
- Impossible to take both pulleys out



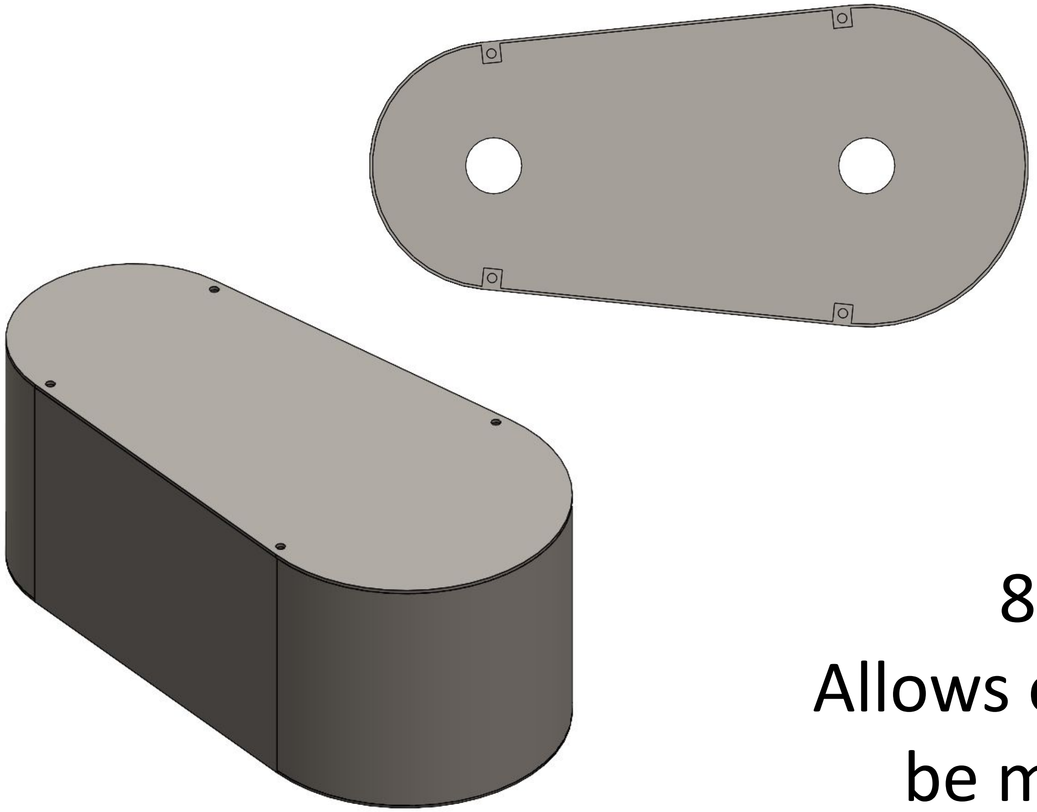




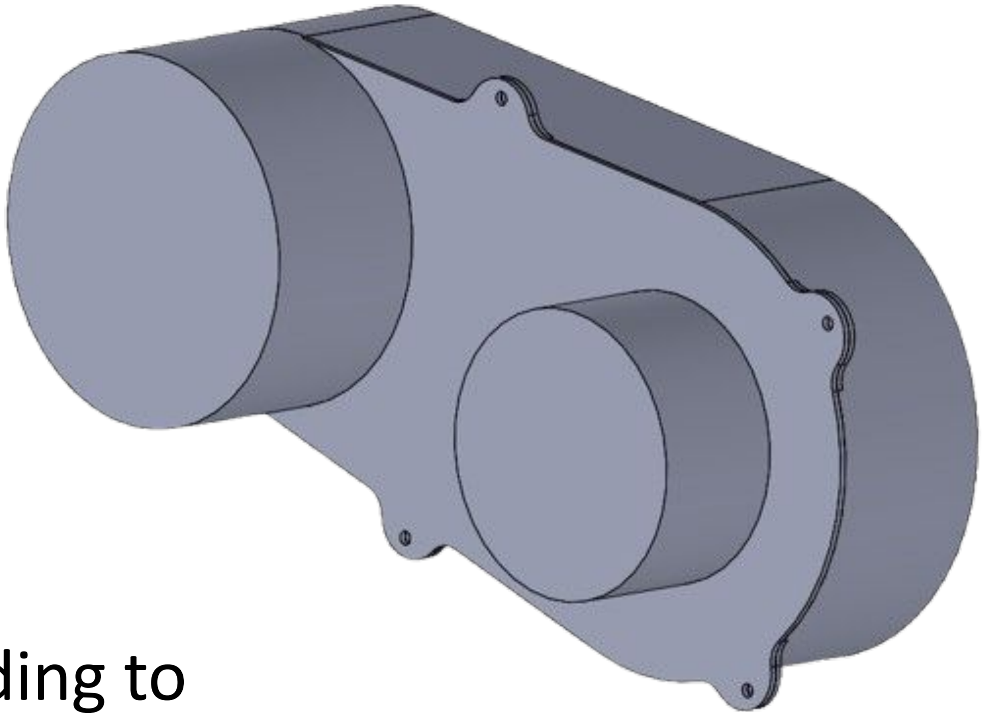


New Design Proposal

Option 1: Large Profile Guarding



Option 2: Smaller Profile Guarding



8.5in CVT!
Allows our guarding to be much smaller

Materials in Use

Between steel and aluminum, we decided to use **steel** because it allows it to be thinner and we don't want to weld aluminum (it sucks).

AISI 1010 steel - 16 gauge (1.519 mm)

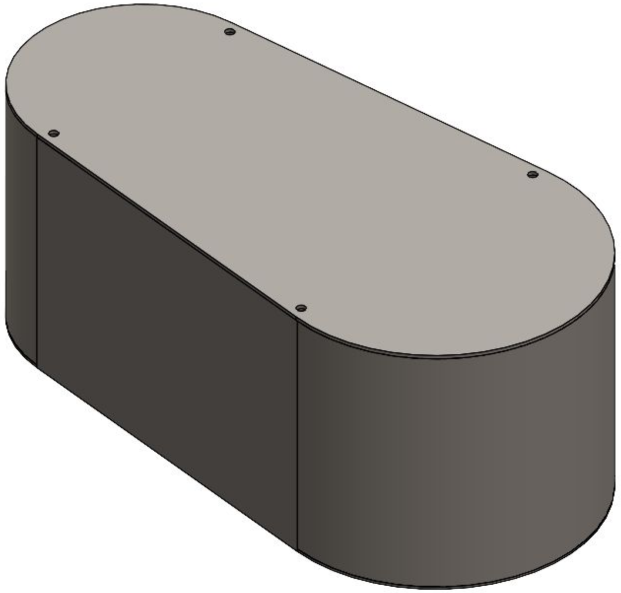
- easier to bend than 14 gauge (1.897 mm)
- meets rule requirements
- easily machinable

AISI 1011 steel

- stronger than AISI 1010 steel
- not meant for cold work

Material selection ultimately falls on **availability** as most common AISI steel types (>=1010) meet rule requirements.

How the CVT Guarding Fits into the Car

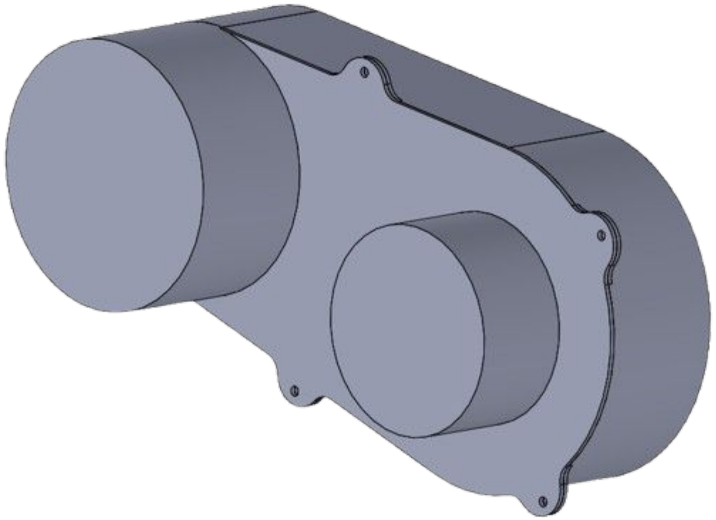


Pros: Easier to Manufacture

Doesn't have the mounting tabs of two sections

Cons: Big profile

Potentially unable to fit into design depending on car layout



Pros: Smaller profile

Adaptable to a lot of changes

Cons: More difficult to manufacture

Has extra tabs

Thank you!
Questions?